If I Invest Tomorrow: A Validated Financial Decision Simulator with a Tool-Grounded AI Agent for Indian Markets

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Abstract—Retail investors are usually given either a single price forecast or a static allocation, without the probability of reaching their goal or the loss they should expect in a crash. We present If I Invest Tomorrow, an end-to-end decision simulator for Indian markets that turns an investor's amount, horizon, risk tolerance, target return and preferred sectors into a concrete plan for the next trading day, in rupees and whole shares. The engine combines a CAPM and history prior for expected returns, a fixed cross-sectional price-pattern signal, a Ridge model for next-month volatility, a Gaussian-mixture market-regime model, Isolation-Forest anomaly detection, five optimised strategies, a regime-switching Student-t Monte Carlo simulation and nine stress scenarios. A locally hosted language model acts as an agent that calls 29 engine tools and manages a demo portfolio under rules enforced in code. All models are evaluated with an expanding-window walk-forward design on 32 assets (2014–2026): model choices are made on 2017–20 and reported on 2021–26. The volatility model reaches an out-of-sample R^2 of 0.604 on log volatility against 0.503 for the best naive rule; the return signal has an information coefficient of 0.057 and an out-of-sample R^2 of +0.22%, small but positive where fitted models are negative. In a monthly walk-forward backtest with realistic NSE trading costs the full system reaches a Sharpe ratio of 0.80 against 0.36 for a NIFTY 50 ETF, and over 2017–26 it beats the ETF in every two-year period. We also report what does not hold: the return signal's net contribution is not distinguishable from zero, the deflated Sharpe ratio is convincing over 2017–26 but only suggestive on 2021–26 alone, one-year Monte Carlo bands held 84% of outcomes at a nominal 90% (1-month and 3-month bands hold 89–90%), and the system's edge is lower risk rather than higher return.

Index Terms—portfolio optimisation, Monte Carlo simulation, volatility forecasting, walk-forward validation, market regimes, LLM agents, robo-advisory, Indian equity markets

I. Introduction

A retail investor deciding what to buy tomorrow faces three questions: what to buy, how much of each asset, and how likely the plan is to reach a target return and how much it could lose in a crash. Price-prediction tools answer none of these well: monthly stock returns are close to unpredictable [7], so a single forecast says little about outcomes. Robo-advisors provide allocations, usually mean–variance based [21], but rarely the probability of meeting a goal or a stress test. Finance chatbots built on large language models can explain concepts, yet tend to invent numbers and cannot act safely on an account.

This paper describes If I Invest Tomorrow, a decision simulator for Indian markets that rests the decision on risk and probability rather than on a point forecast. Its contributions are:

1) an integrated pipeline from cleaned NSE data through return and volatility models, five optimised strategies, a regime-switching Monte Carlo simulation and stress tests to a whole-share plan for the next trading day;
2) an honest validation protocol: expanding-window walk-forward evaluation with purged targets, model choices made on 2017–20 under rules fixed before each run, and results reported on 2021–26, including what failed;
3) a calibration analysis of the Monte Carlo against 104 real one-year outcomes and a robustness analysis of the portfolio results over 2017–26 with realistic Indian trading costs and block-bootstrap confidence intervals;
4) a tool-grounded AI agent: a local 4-billion-parameter language model that only selects and explains engine tools, with trading permissions and safety checks enforced in code.

II. Related Work

Markowitz [1] introduced mean–variance portfolio selection and the efficient frontier; Sharpe [2] derived the CAPM, which we use as a prior for expected returns. Sample covariance matrices are noisy when assets are many and history is short; the Ledoit–Wolf shrinkage estimator [3] gives a well-conditioned alternative. Rockafellar and Uryasev [4] showed that conditional value-at-risk can be minimised by linear programming, the basis of our crash-resistant strategy. Hamilton [5] modelled economic series

TABLE I
Investable universe (32 assets, daily, 2014–2026)

Class	n	Examples
Stocks	26	RELIANCE, TCS, HDFCBANK, SBIN
Equity ETFs	3	NIFTY, Next 50, Bank NIFTY
Gold	1	GOLDBEES
Government bonds	1	LTGILTBEES
Liquid fund	1	LIQUIDBEES

as switching between latent regimes, and Glasserman [6] surveys Monte Carlo methods in financial engineering.

Gu, Kelly and Xiu [7] compared machine-learning models for predicting US stock returns: the best neural network reaches a monthly out-of-sample R^2 of 0.40% and a regression on all predictors −3.46%. Fischer and Krauss [9] report 54.3% daily direction accuracy for LSTM networks on the most extreme daily picks, before costs. Gradient-boosted trees [8] are common baselines. Documented cross-sectional effects include short-term reversal [10], momentum [11], residual momentum [12] and reversal relative to the industry [13]. Unlike returns, volatility is persistent and forecastable [15]; range-based estimators [16]–[18] use the daily high and low, and Andersen and Bollerslev [19] showed that forecast accuracy looks low when it is measured against a noisy proxy of true volatility. Isolation Forest [20] detects anomalies by how easily points are isolated.

Language-model agents that interleave reasoning with tool calls [22] or learn to call tools [23] reduce made-up answers by delegating facts and computation to tools. Our system follows this pattern but moves every safety decision out of the model.

III. Data

The universe has 32 assets traded on the NSE: 26 large stocks from 12 sectors, three equity ETFs (NIFTY 50, NIFTY Next 50, Bank NIFTY), a gold ETF, a long-dated government-bond ETF and a liquid fund (Table I). Daily open, high, low, close and volume come from Yahoo Finance for 1 October 2014 to 1 October 2026 (2,950 trading days), together with the NIFTY 50 index, India VIX, Brent crude, USD/INR and the US 10-year yield. A longer NIFTY and India VIX history from 2007 is used only by the regime model, so that it learns from both the 2008 and 2020 crashes. Quarterly results (announcement dates, EPS estimates and surprises) of the 26 stocks from about 2005 feed the volatility model's earnings-calendar features.

Cleaning aligns all series to the NSE calendar, fills gaps of at most five days, and detects bad prints. It found and fixed wrong-scale prices in three ETFs (December 2019) and erratic gilt-ETF prints; the liquid fund, whose quoted price ignores its daily payouts, is rebuilt from the RBI repo-rate path.

IV. Methodology

A. Validation protocol

Every model is evaluated by an expanding-window walk-forward: for each calendar year it is refit on data before that year only, with targets purged for 1.5× the forecast horizon so that no training target overlaps the test year. Choices (features, signals, blend weights, simulation settings) are made on 2017–20; from the seventh model-search round onwards the rule for adopting a change was fixed before running: a candidate must beat the model in use on 2017–20. The years 2021–26 are reported. Because these years were also inspected across earlier rounds, we treat them as an evaluation period subject to some selection bias rather than as a fully untouched test.

B. Expected returns

The prior combines the CAPM with each asset's 5-year history:

$$\mu_i^{\text{prior}} = 0.6\,\big(r_f + \beta_i\,\text{ERP}\big) + 0.4\,h_i, \tag{1}$$

with $r_f = 6\%$, β_i from a regression on NIFTY 50 excess returns, h_i the historical annual return clipped to $[-5\%, 30\%]$, and $\text{ERP} = \text{clip}\big(0.5(h_m - r_f) + 0.5 \times 5.5\%,\ 3\%,\ 8\%\big)$.

None of the fitted return models (Ridge, random forest, XGBoost, GRU) showed skill on 2017–20. The production signal is therefore a fixed rule with nothing fitted. From residual returns $e_{i,t} = r_{i,t} - \hat{\beta}_{i,t-1} r_{m,t}$ it ranks, each day, residual momentum m_i (months $t-12$ to $t-1$, risk-scaled) [12], the residual one-month return v_i (reversal) and the residual one-month return relative to the asset's sector $v_i - \bar{v}_{s(i)}$ [13]:

$$s_i = z\Big(\tfrac{1}{3}\big[\,\text{rk}(m_i) - \text{rk}(v_i) - \text{rk}(v_i - \bar{v}_{s(i)})\big]\Big)\,, \tag{2}$$

where rk is the cross-sectional percentile rank and z a cross-sectional z-score. Grinold's refined forecast [14] turns the score into an expected 21-day relative return $f_i = \text{IC}\cdot\sigma_{cs}\cdot s_i$ (IC = 0.061 from 2017–20, σ_{cs} the cross-sectional volatility of the target). The final expected return adds the annualised forecast as a bounded tilt, weighted by the measured skill:

$$\mu_i = \mu_i^{\text{prior}} + \lambda\,\text{clip}\big((f_i - \bar{f})\tfrac{252}{21},\ \pm 15\%\big), \tag{3}$$

with $\lambda = \text{clip}(5\,\text{IC},\ 0,\ 0.5) = 0.29$.

C. Volatility forecasting

The target is the log of the annualised realised volatility of the next 21 daily returns. A Ridge regression ($\alpha = 3000$) uses 62 point-in-time features: realised volatility over 1–252 days, Parkinson, Garman–Klass and Rogers–Satchell range estimators [16]–[18], semivariances, volume, market volatility and India VIX, the asset's 3-year volatility level, seasonal terms (the same window a year earlier, the share of the next month in a results season), an earnings calendar (the projected next results date and the stock's typical

result-day jump, both from past announcements only), and asset-class interactions. The forecast blends the model with last quarter's volatility:

$$\widehat{\log \sigma_i} = 0.85\, \hat{y}_i^{\text{Ridge}} + 0.15\, \log \sigma_i^{(63)}. \tag{4}$$

D. Regimes and anomalies

A three-component Gaussian mixture on market features (returns, volatility, drawdown and VIX) labels each day as Bull/Calm, Neutral/Sideways or Bear/Volatile; a Markov chain of daily transitions gives regime persistence. Over 2007–2026 the Bear state covers 8.6% of days with 45% annual volatility and an average VIX of 42.5. An Isolation Forest [20] flags unusual market days and assets.

E. Strategies and recommendation

Long-only weights are optimised under per-asset caps set by the risk profile and an optional sector filter, using the Ledoit–Wolf covariance [3]. Five strategies are built: Max Return, Min Risk (minimum variance), Max Sharpe, Goal-Based (the frontier point with the highest simulated probability of reaching the target) and Crash-Resistant (minimum CVaR over history and all stress scenarios [4]). Each is simulated and stress-tested; strategies that breach the profile's limits on volatility, worst stress loss or one-year VaR are dropped. Among the rest the rule picks the highest probability of reaching the target, breaking ties within two percentage points by the higher Sharpe ratio. The plan is converted to whole shares at the latest close.

F. Monte Carlo simulation

For a portfolio with expected return μ, volatility σ and beta β, each of 10,000 paths evolves daily ($\Delta t = 1/252$) as

$$\ln V_{t+1} = \ln V_t + \left[\ln(1+\mu+\tau_{k_t}) - \tfrac{1}{2}\sigma_t^2/\Delta t\right]\Delta t + \delta\,\Delta t + \sigma_t \varepsilon_t, \tag{5}$$

where ε_t is Student-t with 5 degrees of freedom scaled to unit variance, k_t follows the fitted Markov chain from today's regime probabilities, $\sigma_t = \sigma\sqrt{\Delta t}\, c_{k_t}\, m$ scales volatility by the regime (c_k) and by a per-path log-normal multiplier m with standard deviation 0.35, and τ_k is the regime's return tilt scaled by β. The per-path drift error $\delta \sim \mathcal{N}(-\tfrac{1}{2}s^2 T, s)$, $s = \sigma/\sqrt{5}$, represents estimation error in a 5-year mean. The simulation outputs the probability of a gain and of reaching the target, percentiles of final value, VaR, CVaR and drawdowns. Nine stress tests apply market shocks of -2%, -5% and -10%, a 25% sector crash, a 100 bp rate shock and a 25% oil shock, and replay the 2020, 2018 and 2021–22 episodes.

V. System Architecture and AI Agent

Fig. 1 shows the architecture. A Python engine implements all models; a FastAPI service exposes it to a Next.js web application with four modules: Planner, Simulator, Markets and Portfolio (a demo wallet of Rs 10 lakh with buy, sell, plan investment, rebalancing and monthly SIPs).

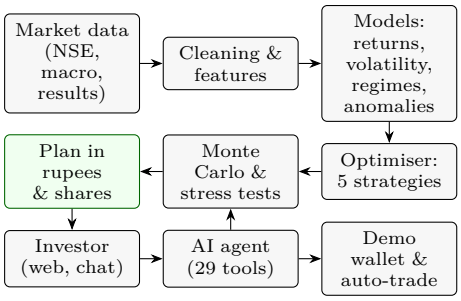


Fig. 1. System architecture. The language model only selects and explains tools; numbers come from the engine and trading rules are enforced in code.

The agent uses Qwen3.5-4B served locally through Ollama. An intent guard first decides which tool a message needs; the model then selects one of 29 tools (plan, simulate, stress test, analyse a stock, trade, rebalance) and explains the result, while the tables and charts shown with each answer come from the tool itself. If the model skips a required tool, the system runs it. Trades are executed only when the user gives an instruction: in ask-first mode orders are staged until the user confirms, in auto-trade mode they are executed all-or-none. An autopilot applies rules after each data update (sell a holding down 15%, trim an oversized position, reduce risk in a crisis regime, invest idle cash). All of these checks are in code, not in the language model; 23 of 23 scripted test conversations pass, and the engine has 84 automated tests.

VI. Results

A. Return prediction

Table II compares our models with published results. Measured as an out-of-sample R^2 against a zero forecast (the Gu–Kelly–Xiu definition), the price-pattern signal reaches $+0.22\%$, while our fitted Ridge model reaches -0.24%, random forest -2.43% and XGBoost -6.46%. An R^2 of $+0.22\%$ means the signal explains 0.22% of the month-to-month variation in returns; it is not a 0.22% return. Its information coefficient on 2021–26 is 0.057 ($t = 2.7$ on independent windows; 0.061, $t = 3.95$, on 2017–20), and it predicts the direction of relative returns correctly 51.5% of the time against 47.5% for the historical mean. Five other documented signals (one-week reversal, the MAX effect, idiosyncratic volatility, abnormal volume, low beta) had no skill on 2017–20, nor did post-earnings drift.

B. Volatility forecasting

On 2021–26 the volatility model reaches $R^2 = 0.604$ on log volatility, against 0.503 for "same as last quarter" and 0.388 for "same as last month"; its average error is 24.6% against 28.6% and 30.6%, and it is ahead of both naive rules in every year (Table III). Within assets (timing only) R^2 is 0.217. Elastic Net, Huber regression, GARCH

TABLE II
Our walk-forward results (2021–26) against published work

Task	Published	Ours
Monthly returns, OOS R^2	NN 0.40%; OLS −3.46% (US stocks) [7]	Signal +0.22%; Ridge −0.24%
Direction	LSTM 54.3%, daily, extreme picks, before costs [9]	51.5% monthly, all 32 assets (mean: 47.5%)
Next-month volatility	Forecastable from past volatility [15]	R^2 0.604 vs 0.503 for the best naive rule
Outcome ranges	No common benchmark	90% band held 84% of 104 outcomes

TABLE III
Volatility forecasts by year: R^2 on log volatility (rows with a full 63-day look-ahead)

Year	Model	Same as last quarter	Same as last month
2021	0.622	0.521	0.440
2022	0.517	0.465	0.338
2023	0.475	0.454	0.274
2024	0.442	0.241	−0.046
2025	0.655	0.581	0.459
2026	0.584	0.274	0.277

and implied-volatility features, gradient-boosted trees and LightGBM did not beat it on 2017–20.

How much higher could R^2 go? The target, the volatility of 21 fat-tailed daily returns, is itself a noisy measurement. Splitting each window into odd and even days gives two independent measurements of the same month; by the Spearman–Brown formula only 84% of the target's variation is true volatility for all assets and 63% for single stocks, which bounds the R^2 of any forecaster. Against a cleaner range-based measure of the same 21 days, our stock forecasts score 0.36 instead of 0.28, the effect described by Andersen and Bollerslev [19].

C. Calibration of the Monte Carlo

At each quarter-end from 2019 to 2025 the full pipeline was rebuilt from data available at that date and a one-year simulation was run for four portfolios (NIFTY ETF, gold, a 60/20/20 balanced mix and eight large stocks), giving 104 real outcomes. The 50/75/90/95/99% bands held 49/74/84/89/100% of them (Fig. 2). Because the outcomes come from overlapping one-year windows of portfolios that move together, we bootstrap whole years of origin dates: the coverage of the 90% band lies between 72% and 94%, so 90% is plausible but not established. The misses are about location rather than width: 8 of the 17 are gold, which beat its expected return by 17% a year in 2019–25, and 7 are the rebound after the 2020 crash; without gold the band held 88%. Earlier periods show the opposite error: on 2016–18 origins the same bands held 98% of outcomes, and on 2009–17 NIFTY origins 100%. Wider bands and recentring on past forecast errors were

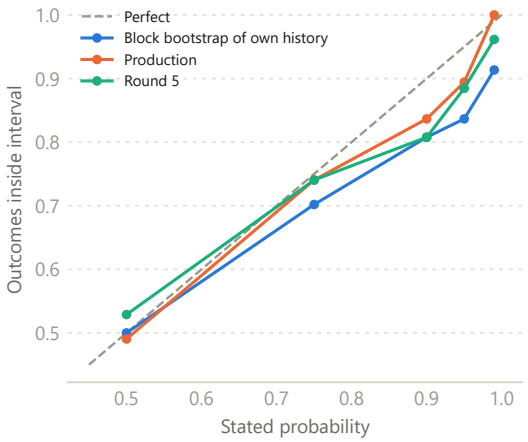


Fig. 2. Calibration of one-year Monte Carlo forecasts (104 outcomes, 2019–25). "Round 5" is the simulator without volatility uncertainty.

TABLE IV
Monthly walk-forward backtest, Jan 2021 – Oct 2026, realistic costs

Strategy	CAGR	Vol.	Sharpe	Max DD	Turn.
Min-variance	11.5%	5.7%	0.95	−7.0%	0.8×
+ volatility model	11.1%	5.4%	0.95	−6.0%	2.7×
Max-Sharpe (prior)	12.2%	11.5%	0.54	−16.7%	3.3×
+ volatility model	14.3%	11.4%	0.73	−14.9%	4.7×
+ return tilt	14.4%	10.4%	0.80	−16.2%	7.5×
+ regime overlay	13.0%	8.6%	0.80	−12.2%	7.4×
NIFTY 50 ETF	10.6%	12.8%	0.36	−16.1%	–
Equal-weight stocks	16.9%	13.0%	0.84	−15.1%	0.7×

tested with the interval score [25], which penalises both misses and width; neither improved it on pre-2019 data, so the simulator was not changed.

D. Portfolio backtest

Every month-end from January 2021 the risk model, volatility model, return signal and regime model were rebuilt from data up to that date, weights were optimised under the medium-risk caps, and trades were made at the close with realistic NSE costs per asset class: securities transaction tax, stamp duty, exchange and SEBI fees with GST, and half the bid–ask spread (about 0.17% to buy and 0.15% to sell a stock, 0.06% for equity ETFs). Table IV and Fig. 3 show the results. In the max-Sharpe family the volatility model lifts the Sharpe ratio from 0.54 to 0.73, the return tilt to 0.80, and the regime overlay keeps 0.80 while cutting the maximum drawdown from −16.2% to −12.2%. The NIFTY 50 ETF reaches 0.36. An equal-weight basket of the 26 stocks (0.84) matches the full system.

E. Robustness

Extending the walk-forward to January 2017 – October 2026 (117 rebalances; 2017–20 were the selection years, so they are not a clean test), the full system beats the NIFTY ETF in every two-year period (Fig. 4). Over 2017–26 its Sharpe ratio is 0.84 against 0.41, and a stationary

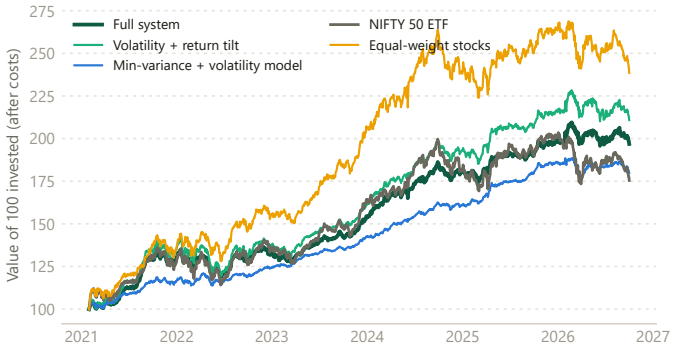


Fig. 3. Growth of 100 invested, after costs, January 2021 to October 2026.

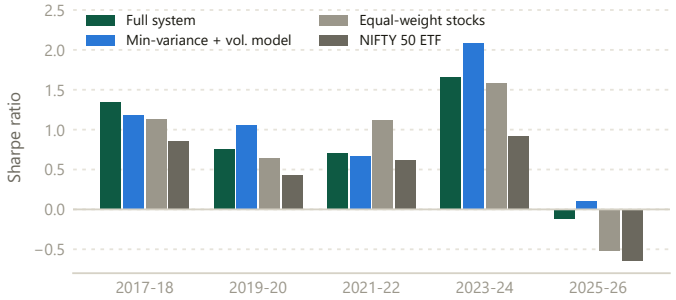


Fig. 4. Sharpe ratio by period, walk-forward 2017–26 with realistic costs (2025–26 runs to October 2026).

block bootstrap [24] of daily returns puts the difference at +0.43 with a 90% interval of +0.01 to +0.83. Doubling the trading costs lowers the full system's 2021–26 Sharpe ratio by 0.11, and the fixed depository charge per security sold costs a Rs 1 lakh account rebalanced monthly up to 0.15. Two results are weaker than the headline. The volatility model's gain in the max-Sharpe family (+0.07 over 2017–26) has a 90% interval that includes zero, and moving the start of the 2021–26 window by one month moves a strategy's Sharpe ratio by about 0.1. The app's five strategies, chosen at 26 to 34 quarter-ends since 2017 and held with quarterly rebalancing, lost far less than the ETF in their worst year (+2% for the Goal-Based plan against −21%), but over three years only Max Return clearly beat it on return; the Goal-Based plan did so in 35% of the windows.

F. How far the evidence goes

Because the 2017–20 years chose the models and the 2021–26 years were inspected during development, we test the headline results further; no model was changed by these tests. First, the return signal: the full system with and without it differs by +0.07 Sharpe on 2021–26 (90% block-bootstrap interval −0.12 to +0.28), +0.04 over 2017–26 and −0.01 on 2017–20, so its net value after costs is not distinguishable from zero. Second, the deflated Sharpe ratio [26], which asks whether the Sharpe ratio exceeds the best that N skill-free trials would produce,

TABLE V
Sensitivity of the recommended plan to simulation assumptions (sampling error of P(target) about ±0.5 points)

Assumption	P(target)	5% worst	Rule picks
Production	53%	92,364	Goal-Based
Normal shocks	53%	92,930	Goal-Based
Student-t, 3 d.o.f.	53%	92,647	Goal-Based
Bootstrap of own history	61%	1,08,371	Goal-Based
Crisis correlations	49%	77,202	Max Sharpe
No regime switching	53%	92,995	Goal-Based
Regimes twice as persistent	53%	91,983	Goal-Based
No volatility uncertainty	53%	94,218	Goal-Based
Volatility uncertainty 0.5	53%	91,184	Goal-Based
No return signal	50%	91,075	Max Return
Expected return −2 points	44%	87,572	Goal-Based
CAPM-only returns	28%	78,810	Goal-Based
History-only returns	80%	1,11,797	Max Return

using the skewness and kurtosis of daily returns and the variance of the Sharpe ratios of the strategy variants actually tested: for the full system it is 0.81–0.89 on 2021–26 alone and 0.95–0.98 over 2017–26, for N between 9 (the portfolio variants) and 300 (every model candidate counted as independent). Third, calibration by horizon on monthly forecast dates: the 90% band holds 89% of 1-month outcomes (interval 85–92%, about 81 independent windows), 90% at 3 months and 85% at one year (75–94%, about 7 windows), so the simulator is well calibrated where outcomes are numerous. Finally, Table V varies the simulation's assumptions for the recommended plan. Tails, regime dynamics and volatility uncertainty barely matter; expected returns and crisis correlations dominate, and under them the rule's recommendation can change. The probability of reaching the target is therefore best stated with an assumption-sensitivity range, 53% (44–61%), which is not a statistical confidence interval.

G. Case study

For an investor with Rs 1,00,000, a three-year horizon, medium risk and a 12% target, the engine recommends the Goal-Based plan: an expected return of 14.0% with 11.6% volatility, a 53% probability of reaching the target, a 92% probability of a gain, a median final value of Rs 1.43 lakh (5th–95th percentile Rs 0.92–2.19 lakh), and survival of 8 of 9 stress tests (Fig. 5). Chosen with data up to October 2023 and held to October 2026, the same strategy returned 55% against 15% for the NIFTY 50 index; the robustness analysis above shows that this window was a favourable one.

VII. Discussion and Limitations

The system's value lies in risk modelling, not in return forecasting. Volatility is forecastable and improves risk-adjusted performance; returns are barely forecastable, so the signal receives only a 29% weight and its net contribution is small. In the deployed planner the volatility forecast reports next-month risk, while plans held for one to five years use the long-run Ledoit–Wolf covariance;

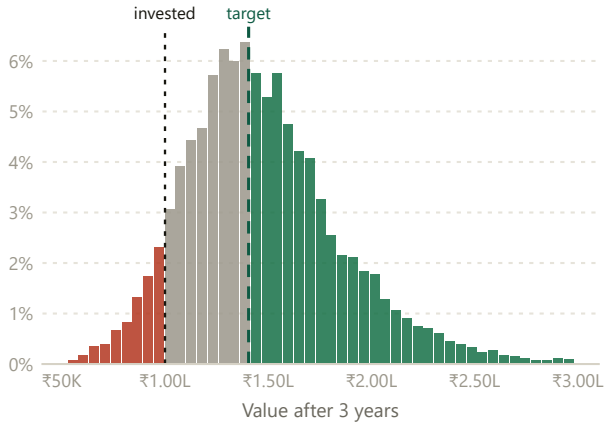


Fig. 5. Simulated value after three years for the sample investor (Rs 1 lakh, Goal-Based plan). Green: target met; grey: gain below target; red: loss.

the monthly-rebalanced backtest uses the forecast to scale the covariance. The main limitations are: (i) the 2021–26 results were inspected over several model-search rounds, so they carry some selection bias; a forward test of the frozen models, which logs each forecast with the code version that produced it, started in October 2026 and is the only untouched evidence; (ii) 12 years contain few independent market regimes, and a five-year Sharpe ratio is sensitive to the start date; (iii) the probability of reaching a target depends on expected-return and correlation assumptions that are hard to estimate; (iv) taxes are not modelled. The demo wallet, auto-trade and plan sizing use the same per-asset trading costs as the backtests. The system is a simulation with demo money and is not investment advice.

VIII. Conclusion

If I Invest Tomorrow turns an investor's profile into a whole-share plan for the next trading day, together with the probability of reaching the goal, the range of outcomes and the loss in a crash, and lets the investor act on it through a tool-grounded AI agent. Its models are validated walk-forward with rules fixed in advance, and the paper reports the weaker findings alongside the strong ones. Future work includes scoring the forward test as new data arrives, taxes, news sentiment, mutual funds and larger local language models.

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